

Cross-sectional study of acral melanoma awareness in a group of South African final phase medical students.

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Acral melanoma refers to melanoma arising on the palms, soles and nail unit, which are sun-protected areas and ultraviolet exposure is not a risk factor. Acral melanoma is associated with a poorer prognosis than other melanoma subtypes most likely due to the high rates of delayed diagnosis. Acral melanoma affects all skin types equally. There is a misconception that people with more pigmented skin types (Fitzpatrick 4-6) do not develop melanoma, due to the protective effect of melanin. The aim of the study was to determine acral melanoma knowledge and awareness of a group of South African, final phase medical students. This was a quantitative and cross-sectional study. A questionnaire consisting of 20 clinical images of skin lesions requiring a diagnosis and management plan was distributed. Responses to six images of melanomas were analysed. Further questions to measure acral melanoma knowledge and related issues were included in the study. A biostatistician appropriately managed statistical analysis. Hundred and one final phase medical students' answers were gathered and analysed. Only 7.9% of the participants diagnosed all six melanomas correctly; 61.4% correctly diagnosed $\geq 50\%$ of the melanomas. While 77.2% of the participants identified all non-acral cutaneous melanoma correctly, only 8.9% identified all acral melanomas. However, of all participants making the correct diagnosis, $>90\%$ selected the appropriate management plan (urgent referral). This study examined a small sample of trainee healthcare workers. The results cannot be assumed to apply to all South African healthcare workers. Responses given in a questionnaire may not reflect actual behaviour. The dermatology division in question has made acral melanoma a research priority, thus acral melanoma knowledge in this group may in fact be better than in other institutions. The present study demonstrates that groups of imminent doctors have low rates of recognition of melanoma, particularly acral melanoma. This is consistent with high levels of primary misdiagnosis of acral melanoma reported in the literature. Fortunately, these participants managed the melanomas they diagnosed appropriately in $>90\%$ of cases. This confirms that the deficit in the participant group is awareness and knowledge. Those aware of the disease immediately acknowledged the need for urgent referral.

TERT amplification but not activation of canonical Wnt/ β -catenin pathway is involved in acral lentiginous melanoma progression to metastasis.

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Acral lentiginous melanoma (ALM) is a rare tumor that occurs on non-sun exposed skin areas of the hands and feet. Reports suggest that ALM exhibits poor prognosis, although mechanisms driving this remain poorly understood. Alterations in TERT and the Wnt/ β -catenin (Wnt) pathway have been suggested to correlate with prognosis of ALM. Thus, immunohistochemical expression of β -catenin and LEF1 along with TERT amplification by FISH was investigated in 34 primary ALMs, 20 metastatic ALMs, 10 primary non-ALMs, and 15 acral nevi. Foot/toe was the most common primary tumor location (85%) for ALM. TERT amplification was detected in 6 of 28 (21.4%) primary ALM, 2 of 8 (25%) primary non-ALM, and 8 of 18 (44.4%) metastatic ALM, the latter showing significantly higher frequency compared with primary melanomas ($P = 0.043$). Most metastatic ALMs positive for TERT amplification lacked BRAF V600E (87.5%). Cytoplasmic and nonnuclear expression of β -catenin was variably detected in all cases. Metastatic ALM revealed lower expression of β -catenin compared with primary ALM ($P = 0.017$). No differences in LEF1 expression were detected among the groups; however, acral nevi showed decreased

labeling with dermal descent, in contrast to melanoma. No molecular-genetic alteration correlated with prognosis. TERT amplification by FISH is a frequent finding in primary ALM and appears to increase in metastatic tumors, suggesting a role in tumor progression to metastasis. Although TERT amplification has been reported to be infrequent in primary non-ALM, it showed comparable frequency with ALM in our series. Our immunohistochemical findings are not fully supportive of activation of either canonical or noncanonical Wnt cascades in ALM. TERT amplification by FISH and LEF1 immunohistochemistry may help in the differential diagnosis between primary ALM and acral nevus. TERT amplification appears to be a promising target for therapy in patients with metastatic ALM.

Malignant melanoma of sun-protected sites: a review of clinical, histological, and molecular features.

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In most cases of cutaneous melanoma, ultraviolet (UV) radiation is recognized as a prominent risk factor. Less is known regarding the mechanisms of mutagenesis for melanoma arising in sun-protected sites, such as acral and mucosal melanoma. Acral and mucosal melanoma share many common features, including a late age of onset, a broad radial growth phase with prominent lentiginous growth, the presence of field cancerization cells, and, in most cases, lack of a precursor nevus. In addition to early chromosomal instability, many of the same genes are also involved in these two distinct melanoma subtypes. To better understand non-UV-mediated pathogenesis in melanoma, we conducted a joint literature review of clinical, histological, and molecular features in acral and mucosal melanoma. We also reviewed the current literature regarding aberrations in KIT, PDGFRA, TERT, and other commonly involved genes. By comparing common features of these two subtypes, we suggest potential mechanisms underlying acral and/or mucosal melanoma and offer direction for future investigations.

Vitamin D receptor immunohistochemistry variability in sun-exposed and non-sun-exposed melanomas.

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Ultraviolet rays are one of the leading factors in the development of melanoma (MM); however, ultraviolet rays seem not to play a role in non-sun-exposed MM, where systemic immunosuppression, anatomical, and physiological features may contribute toward the development of the malignancy. Our aim was to evaluate vitamin D receptor (VDR) expression in shield-site melanoma (ST-MM) and non-shield-site melanoma (NST-MM) to find features that could explain the different biological behavior of MM according to the area of onset. We reviewed 118 specimens of MM. VDR expression was assayed using immunohistochemistry by dividing the specimens according to the anatomical area. We included MM of the soles, feet, hands, gluteus, scrotum, skin of the penile shaft, and large vaginal labia in the ST-MM group. The NST-MM group was divided into two main categories: NST-MM of chronic sun-exposed areas, including MM of the face, scalp, neck, back of the hands, and NST-MM of intermittent sun-exposed areas, including MM of the trunk, lower, and upper limbs. In shield sites, 66.67% of MMs showed VDR expression; in intermittent sun-exposed areas, 33.3% showed VDR expression; and in chronic sun-exposed areas, only 4.66% showed VDR expression. A similar behavior was observed for Breslow's thickness, where VDR staining intensity was higher in thicker lesions, ranging between 60 and 100%. We

found that VDR expression decreased from ST-MM to NST-MM. These findings confirm the hypothesis that different pathways are involved in ST-MM and NST-MM.

Consensus molecular subtypes of colorectal cancer in clinical practice: A translational approach.

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The identification of several genetic mutations in colorectal cancer (CRC) has allowed a better comprehension of the prognosis and response to different antineoplastic treatments. Recently, through a systematic process, consensus molecular subtypes (CMS) have been described to characterize genetic and molecular mutations in CRC patients. Through CMS, CRC patients can be categorized into four molecular subtypes of CRC by wide transcriptional genome analysis. CMS1 has microsatellite instability and mutations in CIMP and BRAF pathways. CMS2, distinguished by mutations in specific pathways linked to cellular metabolism, also has a better prognosis. CMS3 has a KRAS mutation as a hallmark. CMS4 presents mutations in fibrogenesis pathways and mesenchymal-epithelial transition, associated with a worse prognosis. CMS classification can be a meaningful step in providing possible answers to important issues in CRC, such as the use of adjuvant chemotherapy in stage II, personalized first-line chemotherapy for metastatic CRC, and possible new target treatments that address specific pathways in each molecular subtype. Understanding CMS is a crucial step in personalized medicine, although prospective clinical trials selecting patients by CMS are required to pass proof-of-concept before becoming a routine clinical tool in oncology routine care.

CINSARC signature outperforms gold-standard TNM staging and consensus molecular subtypes for clinical outcome in stage II-III colorectal carcinoma.

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The outcome of stage II-III colorectal cancer (CRC) is highly variable and therapeutic choice is currently based on TNM staging with a few additional biomarkers. However, studies show that some stage III patients have a better prognosis than some stage II patients. A promising consensus molecular (CMS) classification with prognostic relevance has been developed, but it is not used in daily practice. Our team developed CINSARC, a 67-gene expression prognostic signature, whose prognostic value has been demonstrated in many cancer types. It is applicable to formalin-fixed, paraffin-embedded (FFPE) blocks using NanoString® technology. We investigated whether it could predict outcome in stage II-III CRC. We established the CINSARC classification on the TCGA retrospective cohort comprising 297 stage II-III CRC patients using RNA sequencing and on a second independent cohort comprising 169 cases using NanoString® technology. We compared its recurrence-free and overall survival prognostic value with TNM staging and CMS classification. In the TCGA cohort, we showed that CINSARC significantly splits the population of stage II-III CRC into two groups with different progression-free interval ($P = 1.68 \times 10^{-2}$; $HR = 1.87$ [1.11-3.16]) and overall survival ($P = 3.73 \times 10^{-3}$; $HR = 2.45$ [1.31-4.59]) and is a strong prognostic factor in multivariate analysis, outperforming TNM staging and CMS classification. We validated these results in the second cohort by applying CINSARC on FFPE samples with Nanostring® technology. CINSARC is a ready-to-use tool with a robust independent prognostic value in stage II-III CRC.

Consensus Molecular Subtypes of Colorectal Cancer and their Clinical Implications.

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The colorectal cancer (CRC) Subtyping Consortium has unified six independent molecular classification systems, based on gene expression data, into a single consensus system with four distinct groups, known as the Consensus Molecular Subtypes (CMS); clinical implications are discussed in this review. This article is based on a literature review relevant to the CMS of CRC indexed in PubMed (US National Library of Medicine) as well as the authors' own published data. The CMS were determined and correlated with epigenomic, transcriptomic, microenvironmental, genetic, prognostic and clinical characteristics. The CMS1 subtype is immunogenic and hypermutated. CMS2 tumors are activated by the WNT- β -catenin pathway and have the highest overall survival. CMS3 feature a metabolic cancer phenotype and CMS4 cancers have the worst survival and have a strong stromal gene signature. The Consensus Molecular Subtypes of CRC may better inform clinicians of prognosis, therapeutic response, and potential novel therapeutic strategies.

Clinical Value of Consensus Molecular Subtypes in Colorectal Cancer: A Systematic Review and Meta-Analysis.

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The consensus molecular subtypes (CMSs) of colorectal cancer (CRC) capture tumor heterogeneity at the gene-expression level. Currently, a restricted number of molecular features are used to guide treatment for CRC. We summarize the evidence on the clinical value of the CMSs. We systematically identified studies in Medline and Embase that evaluated the prognostic and predictive value of CMSs in CRC patients. A random-effect meta-analysis was performed on prognostic data. Predictive data were summarized. In local disease, CMS4 tumors were associated with worse overall survival (OS) compared with CMS1 (hazard ratio [HR] = 3.28, 95% confidence interval = 1.27 to 8.47) and CMS2 cancers (HR = 2.60, 95% confidence interval = 1.93 to 3.50). In metastatic disease, CMS1 consistently had worse survival than CMS2-4 (OS HR range = 0.33-0.55; progression-free survival HR range = 0.53-0.89). Adjuvant chemotherapy in stage II and III CRC was most beneficial for OS in CMS2 and CMS3 (HR range = 0.16-0.45) and not effective in CMS4 tumors. In metastatic CMS4 cancers, an irinotecan-based regimen improved outcome compared with oxaliplatin (HR range = 0.31-0.72). The addition of bevacizumab seemed beneficial in CMS1, and anti-epidermal growth factor receptor therapy improved outcome for KRAS wild-type CMS2 patients. The CMS classification holds clear potential for clinical use in predicting both prognosis and response to systemic therapy, which seems to be independent of the classifier used. Prospective studies are warranted to support implementation of the CMS taxonomy in clinical practice.

Principles of Molecular Utility for CMS Classification in Colorectal Cancer Management.

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Colorectal cancer (CRC) is the second cause of cancer-related deaths in both sexes globally and presents different clinical outcomes that are described by a range of genomic and epigenomic alterations. Despite the advancements in CRC screening plans and treatment strategies, the prognosis of CRC is dismal. In the last two decades, molecular biomarkers predictive of prognosis have been identified in CRC, although

biomarkers predictive of treatment response are only available for specific biological drugs used in stage IV CRC. Translational clinical trials mainly based on "omic" strategies allowed a better understanding of the biological heterogeneity of CRCs. These studies were able to classify CRCs into subtypes mainly related to prognosis, recurrence risk, and, to some extent, also to treatment response. Accordingly, the comprehensive molecular characterizations of CRCs, including The Cancer Genome Atlas (TCGA) and consensus molecular subtype (CMS) classifications, were presented to improve the comprehension of the genomic and epigenomic landscapes of CRCs for a better patient management. The CMS classification obtained by the CRC subtyping consortium categorizes CRC into four consensus molecular subtypes (CMS1-4) characterized by different prognoses. In this review, we discussed the CMS classification in different settings with a focus on its relationships with precursor lesions, tumor immunophenotype, and gut microbiota, as well as on its role in predicting prognosis and/or response to pharmacological treatments, as a crucial step towards precision medicine.

Consensus molecular subtypes (CMS) in metastatic colorectal cancer - personalized medicine decision.

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Background Colorectal cancer (CRC) is one of the most common types of cancer in the world. Metastatic disease is still incurable in most of these patients, but the survival rate has improved by treatment with novel systemic chemotherapy and targeted therapy in combination with surgery. New knowledge of its complex heterogeneity in terms of genetics, epigenetics, transcriptomics and microenvironment, including prognostic and clinical characteristics, led to its classification into various molecular subtypes of metastatic CRC, called consensus molecular subtypes (CMS). The CMS classification thus enables the medical oncologists to adjust the treatment from case to case. They can determine which type of systemic chemotherapy or targeted therapy is best suited to a specific patient, what dosages are needed and in what order. Conclusions CMS in metastatic CRC are the new tool to include the knowledge of molecular factors, tumour stroma and signalling pathways for personalized, patient-orientated systemic treatment in precision medicine.